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Information processing apparatus for acquiring actual viewpoint position and orientation and virtual viewpoint position and orientation of user, information processing method, and storage medium

Abstract

An information processing apparatus includes a first acquisition unit for acquiring an actual viewpoint position and orientation and a virtual viewpoint position and orientation of a first user, a second acquisition unit for acquiring an actual viewpoint position and orientation and a virtual viewpoint position and orientation of a second user, and a correction unit for correcting the virtual viewpoint position and orientation of the first user based on the virtual viewpoint position and orientation of the second user and a relative position and orientation between the actual viewpoint position and orientation of the first user and the actual viewpoint position and orientation of the second user.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

(1) The present disclosure relates to an information processing technique for providing, for example, virtual reality, mixed reality, and augmented reality.

Description of the Related Art

(2) Virtual reality, mixed reality, and augmented reality enable a user to observe various locations, such as buildings, represented by virtual three-dimensional (3D) models by moving only a virtual viewpoint to draw a virtual 3D model according to a user operation, without the user moving by itself. The function of moving the virtual viewpoint according to the user operation is referred to as a walk-through function. Virtual reality is often abbreviated as “VR”. Mixed reality is often abbreviated as “MR”. Augmented reality is often abbreviated as “AR”.

(3) To smoothly use the walk-through function, it is important to make clear the positional relationship between the user and the virtual 3D model to be observed and to easily recognize what the user is watching from where. For a plurality of users to observe a 3D model or the like and to communicate with each other, it is important for each user to recognize the positional relationship between the user and the other users without any confusion. Japanese Patent Application Laid-Open No. 2006-79174 discusses a technique having an actual-size observation mode in which an observer observes a virtual 3D space by being immersed in the virtual 3D space in an actual size,

and a reduced-size observation mode in which the observer observes a virtual 3D space by reducing the size of a virtual 3D space computer graphics (CG) image. According to this technique, a CG map obtained by adding position and orientation information about the user is presented in the reduced-size observation mode.

SUMMARY OF THE DISCLOSURE

(4) According to an aspect of the present disclosure, an information processing apparatus includes a first acquisition unit for acquiring an actual viewpoint position and orientation and a virtual viewpoint position and orientation of a first user, a second acquisition unit for acquiring an actual viewpoint position and orientation and a virtual viewpoint position and orientation of a second user, and a correction unit for correcting the virtual viewpoint position and orientation of the first user based on the virtual viewpoint position and orientation of the second user and a relative position and orientation between the actual viewpoint position and orientation of the first user and the actual viewpoint position and orientation of the second user.

(5) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram illustrating a configuration example of an information processing apparatus according to a first exemplary embodiment.

(2) FIG. 2 illustrates a relationship between a virtual viewpoint and an actual viewpoint by walk-through.

(3) FIG. 3 illustrates an example of a mixed reality image viewed from a viewpoint of a user.

(4) FIG. 4 is a flowchart illustrating an information processing flow according to the first exemplary embodiment.

(5) FIG. 5 illustrates a mixed reality image viewed from the virtual viewpoint of the user after correction.

(6) FIG. 6 is a block diagram illustrating a configuration example of an information processing apparatus according to a second exemplary embodiment.

(7) FIG. 7 is a flowchart illustrating an information processing flow according to the second exemplary embodiment.

(8) FIG. 8 is a block diagram illustrating a configuration example of an information processing apparatus according to a third exemplary embodiment.

(9) FIG. 9 is a flowchart illustrating an information processing flow according to the third exemplary embodiment.

(10) FIG. 10 illustrates a hardware configuration example of the information processing apparatus.

DESCRIPTION OF THE EMBODIMENTS

(11) Exemplary embodiments of the present disclosure will be described below with reference to the drawings. The following exemplary embodiments are not intended to limit the present disclosure. Not all of the combinations of the features described in the exemplary embodiments are indispensable for the solution to the issues of the present disclosure. The components of the exemplary embodiments can be appropriately modified or changed depending on the specifications of the apparatus to which the present disclosure is applied and various conditions (e.g., usage conditions, and usage environment). Some of the exemplary embodiments to be described below may be combined as appropriate. In the following exemplary embodiments, the same components are denoted by the same reference numerals.

(12) FIG. 1 is a block diagram illustrating a configuration example of a system including an information processing apparatus 319 according to a first exemplary embodiment. The system

according to the present exemplary embodiment is a system capable of providing a user with, for example, virtual reality (VR), mixed reality (MR), and augmented reality (AR). The present exemplary embodiment illustrates an example of providing MR to the user. The system according to the present exemplary embodiment also provides a walk-through function that enables the user to observe various locations, such as buildings, represented by virtual three-dimensional (3D) models by moving a virtual viewpoint to draw a virtual 3D model according to a user operation, without the need for the user to move by the user itself.

(13) Before the description of the configuration and operation of the system according to the present exemplary embodiment, a positional relationship between a viewpoint position of each user and a 3D model or the like in a MR space in a case where a plurality of users is observing the 3D model and a certain user uses the walk-through function will be described with reference to FIGS. 2 and 3.

(14) FIG. 2 schematically illustrates the layout of positions of the plurality of users and the 3D model, and the like in the mixed reality space, and the virtual viewpoint position and the like of a specific user in a case where the specific user executes walk-through in the mixed reality space. In this example, assume that there are two users.

(15) In the example illustrated in FIG. 2, a virtual model **105** and a virtual model **106** are 3D models (3D objects) located in the mixed reality space. A user **101** and a user **102** exist as entities in an actual space. The user **101** and the user **102** are respectively wearing terminal head-mounted displays (hereinafter referred to as HMDs **108**). Assume that the user **101** and the user **102** are observing the mixed reality space in which the virtual model **105** and the virtual model **106** are located through the HMDs **108** that are respectively worn by the user **101** and the user **102**. Also, assume that the user **101** and the user **102** respectively have operation devices **107** to input various instructions in their hands. Also, assume that each of the user **101** and the user **102** can issue an instruction to, for example, move the virtual viewpoint in the mixed reality space through an operation input to the operation device **107**. In the present exemplary embodiment, examples of the instruction input through the operation device **107** include an instruction to execute walk-through and move the virtual viewpoint, and an instruction to input a predetermined trigger operation to be described below.

(16) In this example, assume that the user **102** executes walk-through to move the virtual viewpoint of the user **102** to the position and orientation indicated at a virtual viewpoint **104** illustrated in FIG. 2. In other words, the virtual viewpoint **104** of the user **102** is moved to an intermediate position between the virtual model **105** and the virtual model **106** on the opposite side of the virtual model **105** and the virtual model **106**, and the virtual viewpoint **104** observes the virtual model **105** and the virtual model **106** from the back side thereof. In this case, an avatar **103** representing the user **102** in the position and orientation of the virtual viewpoint **104** is also located (displayed) in the mixed reality space. In this example, the virtual viewpoint **104** of the user **102** indicates the position and orientation in which the user **102** faces the front side and is located on the back side at the intermediate position between the virtual model **105** and the virtual model **106** by walk-through. Thus, the avatar **103** of the user **102** faces the front side.

(17) FIG. 3 schematically illustrates the mixed reality space observed from the viewpoint of the user **101**.

(18) The positional relationship with the user **102** in the MR space and the layout relationship between the virtual models **105** and **106** correspond to those of the example illustrated in FIG. 2. Specifically, in this example, the entity of the user **102** and the actual viewpoint (actual viewpoint of the head) of the user **102** are present on the front left side of the user **101**, and the virtual viewpoint **104** of the user **102** is located at the intermediate position between the virtual models **105** and **106** on the opposite side of the virtual models **105** and **106**. Accordingly, the user **101** can see the entity of the user **102** and the virtual model **105** on the left side, a part of the virtual model **106** on the right back side, and the avatar **103** on the back side at the intermediate position between

the virtual model **105** and the virtual model **106** along with the walk-through of the user **102**.

(19) In the example illustrated in FIGS. **2** and **3**, the way the user **101** can see the mixed reality space is different from the way the user **102** can see the mixed reality space. Specifically, the user **101** can see the mixed reality space as illustrated in FIG. **3**, while the mixed reality image viewed from the virtual viewpoint **104** is displayed on the HMD **108** of the user **102**. Accordingly, the way the user **102** can see the virtual model **105** and the virtual model **106** greatly differs from the way the user **101** can see the virtual model **105** and the virtual model **106**. This makes it difficult for the user **101** to accurately recognize the situation seen by the user **102**. Specifically, in a case where a plurality of users is observing the mixed reality space and a specific user executes walk-through, it is difficult for the other users to accurately recognize the relationship between the virtual viewpoint of each of the other users, and the virtual viewpoint and the actual viewpoint of the specific user during the walk-through. As a result, the users are more likely to be confused in communication.

(20) Therefore, the information processing apparatus **319** according to the present exemplary embodiment is configured to perform processing for correcting a mismatch between the virtual viewpoint of a specific user by walk-through and the virtual viewpoint of each of the other users. The information processing apparatus **319** according to the present exemplary embodiment corrects the virtual viewpoint position and orientation of the other users in order to match the virtual viewpoint position and orientation of the specific user that has moved by walk-through.

(21) If a plurality of users observes the MR space, the system according to the present exemplary embodiment illustrated in FIG. **1** uses an apparatus including the information processing apparatus **319**, an operation unit **305**, and a terminal **302** for each user. An apparatus **301** and an apparatus **318** illustrated in FIG. **1** have the same configuration and include similar processing functions. The apparatus **301** and the apparatus **318** each include the information processing apparatus **319**, the operation unit **305**, and the terminal **302**. In the present exemplary embodiment, assume that a first user in the plurality of users (all provided with mixed reality) observing the mixed reality space uses the apparatus **301**, and a second user that is different from the first user of the apparatus **301** uses the apparatus **318**. In the system according to the present exemplary embodiment, the apparatus **301** and the apparatus **318** operate in cooperation through a transmission unit **315** and a reception unit **314**, which are mounted in the apparatus **301**, and a transmission unit **316** and a reception unit **317**, which are mounted in the apparatus **318**, thereby enabling the users to observe 3D images and the like in the mixed reality space. FIG. **1** illustrates a configuration example in which there are two users and two apparatuses, which are the apparatus **301** and the apparatus **318**, operate in cooperation. However, if there are three or more users, three or more apparatuses operate in cooperation, thereby making it possible to provide the mixed reality to the three or more users. If there is a need to distinguish the user of the apparatus **301** from the user of the apparatus **318**, particularly, in the configuration example of the apparatus **301** to be described below, the first user of the apparatus **301** is referred to as the “user” and the second user of the apparatus **318** is referred to as the “other user”.

(22) The terminal **302** includes an image capturing unit **303** that captures an image of an actual space (actual environment) near each user, and a display unit **304** that presents mixed reality images to each user. In the present exemplary embodiment, the HMD having a configuration in which the image capturing unit **303** and the display unit **304** are mounted in a head-mounted housing is used as the terminal **302**. However, any other type of terminal, such as a tablet terminal or a smartphone terminal, can also be used. The HMD that is described by way of example in the present exemplary embodiment is a video see-through HMD that displays a virtual 3D model superimposed on an actual image captured by the image capturing unit **303**, but instead a so-called optical see-through HMD can be used. The image capturing unit **303** includes a camera for capturing a moving image to be presented to each user, and a camera for acquiring images to be used to estimate the position and orientation of the terminal **302**. Alternatively, the image capturing unit **303** may include one camera having both the function for capturing a moving image to be

presented to each user and the function for acquiring images to be used to estimate the position and orientation. Depending on the method of acquiring the position and orientation and the method of displaying a 3D image superimposed on an actual image, one or both of the camera for capturing a moving image to be presented to each user and the camera for acquiring images to be used to estimate the position and orientation can be omitted. In the present exemplary embodiment, images to be presented to each user can be stereoscopic. Accordingly, in the terminal **302** according to the present exemplary embodiment, the camera for capturing images to be presented to each user and a display device to present mixed reality images are individually mounted for each of the left and right eyes of the user to realize the stereoscopic display. However, the configuration for realizing the stereoscopic display may be changed, as needed, depending on the configuration of the terminal **302**, the method of acquiring the position and orientation of the terminal **302**, or the method of displaying a virtual 3D image superimposed on an actual image. Image information about the actual space (actual environment) captured by the image capturing unit **303** is sent to the information processing apparatus **319**. A mixed reality space image to be displayed on the display unit **304** is generated by the information processing apparatus **319** and is input to the display unit **304**.

(23) The operation unit **305** is an operation device to be when the user inputs various instructions.

(24) The input to the operation unit **305** is implemented by, for example, one or more of a button input from a controller device or the like, a gesture input, and an audio input. The operation unit **305** may be mounted on the terminal **302**, or may be configured as a separate device. While the user can input various instructions through the operation unit **305**, the present exemplary embodiment illustrates an example where an operation for instructing to move the virtual viewpoint of the user is input. In the present exemplary embodiment, the input of the operation for instructing to move the virtual viewpoint of the user includes an input of an operation for moving the virtual viewpoint of the user by using the walk-through function of the apparatus **301**, and an input of a predetermined trigger operation for instructing to move the virtual viewpoint of the user depending on the virtual viewpoint of the other user. Assume that the predetermined trigger operation is input to issue an instruction to move the virtual viewpoint of the user in the apparatus **301** to match the virtual viewpoint of the other user by walk-through executed by the apparatus **318** of the other user. In other words, the trigger operation to move the virtual viewpoint of the user to match the virtual viewpoint of the other user is an operation used as a trigger to execute processing for correcting a mismatch between the virtual viewpoint of the other user by walk-through and the virtual viewpoint of the user. In the present exemplary embodiment, the trigger operation is a trigger to move the virtual viewpoint of the user to match the virtual viewpoint of the other user by walk-through, and is thus hereinafter referred to as a “movement trigger”.

(25) The information processing apparatus **319** includes not only the function for estimating the position and orientation of the terminal **302** and the function for generating the mixed reality space image, but also a function for sharing the actual viewpoint position and orientation and the virtual viewpoint position and orientation between the apparatus **301** of the user and the apparatus **318** of the other user as described below. The information processing apparatus **319** includes an operation acquisition unit **306**, a movement determination unit **307**, an image acquisition unit **308**, an actual viewpoint acquisition unit **309**, a virtual viewpoint generation unit **310**, a combining unit **311**, a correction unit **312**, a relative relationship calculation unit **313**, the reception unit **314**, and the transmission unit **315**. FIG. 1 illustrates only the functional units that are mainly associated with information processing according to the present exemplary embodiment in all the components that are typically included in the information processing apparatus **319** that provides the mixed reality to each user, and the illustration of components other than the functional units is omitted.

(26) The image acquisition unit **308** acquires an actual environment captured image captured by the image capturing unit **303** of the terminal **302**.

(27) The actual viewpoint acquisition unit **309** receives the actual environment captured image

from the image acquisition unit **308**, and calculates the actual viewpoint position and orientation of the user, who is wearing the terminal **302**, based on the captured image. The actual viewpoint acquisition unit **309** sends information about the calculated actual viewpoint position and orientation of the user to the virtual viewpoint generation unit **310**.

(28) If a user operation is performed on the operation unit **305**, the operation acquisition unit **306** acquires operation information corresponding to the user operation. In the present exemplary embodiment, the operation acquisition unit **306** includes a function for acquiring, as operation information, an input of an operation to, for example, move the virtual viewpoint position and orientation, from the user through the operation unit **305**. Examples of the operation for instructing to move the virtual viewpoint position and orientation include an operation for instructing to move the virtual viewpoint of the user by walk-through, and an operation instruction using the movement trigger to move the virtual viewpoint of the user to match the virtual viewpoint of the other user by walk-through. The operation information acquired by the operation acquisition unit **306** is sent to the movement determination unit **307**. The operation information is also sent to the virtual viewpoint generation unit **310**, the correction unit **312**, and the relative relationship calculation unit **313** through the movement determination unit **307**.

(29) The movement determination unit **307** determines whether the operation information acquired by the operation acquisition unit **306** is information indicating the operation for instructing to move the virtual viewpoint of the user. In the present exemplary embodiment, the movement determination unit **307** determines whether the operation information is information indicating an operation to move the virtual viewpoint of the user by walk-through or the like in the apparatus **301**, or information indicating an operation of the movement trigger to move the virtual viewpoint of the user to match the virtual viewpoint of the other user by walk-through. In other words, the movement determination unit **307** functions as a trigger detection unit to detect the movement trigger, and upon detecting the movement trigger, outputs the operation information in which a flag indicating that the movement trigger is input is ON. The flag indicating that the movement trigger is input is hereinafter referred to as a “movement flag”. The operation information obtained after the movement determination processing performed by the movement determination unit **307** is sent to the virtual viewpoint generation unit **310**, the correction unit **312**, and the relative relationship calculation unit **313**.

(30) The virtual viewpoint generation unit **310** receives the actual viewpoint position and orientation information from the actual viewpoint acquisition unit **309**, and receives the operation information through the movement determination unit **307**. The virtual viewpoint generation unit **310** generates the virtual viewpoint position and orientation of the user based on the actual viewpoint position and orientation information and the operation information corresponding to the viewpoint position input by the user. If walk-through or the like is executed in the apparatus **301** to move the virtual viewpoint of the user, the virtual viewpoint position and orientation generated by the virtual viewpoint generation unit **310** corresponds to the virtual viewpoint position and orientation of the user by walk-through or the like. The virtual viewpoint generation unit **310** sends the generated virtual viewpoint position and orientation information about the user and the actual viewpoint position and orientation information about the user received from the actual viewpoint acquisition unit **309**, to the transmission unit **315**. The virtual viewpoint position and orientation information and the actual viewpoint position and orientation information about the user are also sent to the relative relationship calculation unit **313**.

(31) The transmission unit **315** transmits the virtual viewpoint position and orientation information and the actual viewpoint position and orientation information received from the virtual viewpoint generation unit **310**, to the reception unit **317** of the apparatus **318** having the same configuration as the apparatus **301**.

(32) The reception unit **314** receives the actual viewpoint position and orientation information and the virtual viewpoint position and orientation information about the other user. The actual

viewpoint position and orientation information and the virtual viewpoint position and orientation information about the other user are acquired and generated in the same manner as described above in the apparatus **318** having the same configuration as the apparatus **301** and are transmitted from the transmission unit **316**. The actual viewpoint position and orientation information and the virtual viewpoint position and orientation information about the other user received by the reception unit **314** are sent to the relative relationship calculation unit **313**.

(33) Thus, the actual viewpoint position and orientation and the virtual viewpoint position and orientation of the both users are shared between the apparatus **301** and the apparatus **318**.

(34) If the movement flag in the operation information is ON, the relative relationship calculation unit **313** calculates a relative relationship between the actual viewpoint position and orientation of the user and the actual viewpoint position and orientation of the other user, based on the actual viewpoint position and orientation information about the user and the actual viewpoint position and orientation information about the other user received from the reception unit **314**. In the present exemplary embodiment, the relative relationship between the actual viewpoint position and orientation of the user and the actual viewpoint position and orientation of the other user is referred to as “actual viewpoint relative position and orientation”. The relative relationship calculation unit **313** sends, to the correction unit **312**, information about the calculated actual viewpoint relative position and orientation information, the virtual viewpoint position and orientation information about the other user, and the virtual viewpoint position and orientation information about the user. While the present exemplary embodiment illustrates an example where the relative relationship calculation unit **313** calculates the actual viewpoint relative position and orientation when the movement flag is ON, the relative relationship calculation unit **313** may constantly calculate the actual viewpoint relative position and orientation, regardless of whether the movement flag is ON or OFF.

(35) If the movement flag in the operation information is ON, the correction unit **312** executes processing for correcting a mismatch between the virtual viewpoint of the other user and the virtual viewpoint of the user based on the actual viewpoint relative position and orientation received from the relative relationship calculation unit **313** and the virtual viewpoint position and orientation of the other user. The present exemplary embodiment assumes that processing for correcting a mismatch between the virtual viewpoint of the other user by walk-through and the virtual viewpoint of the user is performed. As described in detail below, the correction unit **312** adds the actual viewpoint relative position and orientation between the user and the other user to the virtual viewpoint position and orientation of the other user, thereby correcting the virtual viewpoint position and orientation of the user in order to match the virtual viewpoint of the other user by walk-through. The correction unit **312** sends the virtual viewpoint position and orientation information about the user obtained after the correction processing, to the combining unit **311**. If the movement flag is OFF, the correction unit **312** does not perform the above-described correction processing, and the virtual viewpoint position and orientation information about the user generated by the virtual viewpoint generation unit **310** is sent to the combining unit **311** from the relative relationship calculation unit **313** through the correction unit **312**.

(36) The combining unit **311** generates a 3D image based on the received virtual viewpoint position and orientation, and combines the generated 3D image with the actual environment captured image acquired by the image acquisition unit **308** by superimposing the 3D image on the actual environment image, thereby generating a mixed reality image as a combination of the virtual image and the actual environment image. In this case, if the movement flag is ON, the mixed reality image generated by the combining unit **311** corresponds to the image that can be seen in the virtual viewpoint position and orientation of the other user by walk-through. In contrast, if walk-through or the like is executed in the apparatus **301** of the user and the movement flag is OFF, the mixed reality image generated by the combining unit **311** corresponds to the image that can be seen in the virtual viewpoint position and orientation along with the walk-through or the like of the user.

(37) FIG. 4 is a flowchart illustrating an information processing flow in which the actual viewpoint position and orientation and the virtual viewpoint position and orientation of the both users are shared between the information processing apparatus **319** of the apparatus **301** and the apparatus **318** of the other user as described above, and in which the mixed reality image is generated depending on whether the movement trigger is detected. While FIG. 4 illustrates only the processing performed in the information processing apparatus **319** of the apparatus **301**, the same information processing is also performed in an information processing apparatus of the apparatus **318**.

(38) In step **S410**, the actual viewpoint acquisition unit **309** acquires the actual viewpoint position and orientation of the user in world coordinates of the actual space. As a method for acquiring the actual viewpoint position and orientation, for example, a method can be used of capturing an image of a marker located in the actual space and estimating the actual viewpoint position and orientation based on the layout of feature points of the marker in the captured image. For example, simultaneous localization and mapping (SLAM) can also be used of simultaneously performing self-position estimation and creation of an environmental map using natural feature points in an actual image obtained by capturing an image of the actual space. Alternatively, an external measurement apparatus, such as a motion capturing apparatus, may be used to acquire the actual viewpoint position and orientation. The origin and orientation of the world coordinate system representing the actual space are assumed to determine in advance using a marker or a specific real object. In the case of using any one of the above-described methods for acquiring the actual viewpoint position and orientation, if the apparatus **301** and the apparatus **318** use different methods, an error may occur in the acquisition result. It may thus be desirable for the apparatus **301** and the apparatus **318** to use the same method.

(39) In step **S420**, the movement determination unit **307** determines whether the movement trigger to issue an instruction to move the virtual viewpoint of the user to match the virtual viewpoint of the other user by walk-through in the apparatus **318** is detected in the operation information received from the operation acquisition unit **306**. If the movement trigger is detected, the movement determination unit **307** outputs the operation information in which the movement flag is ON.

(40) In step **S430**, the virtual viewpoint generation unit **310** calculates the virtual viewpoint position and orientation of the user based on the operation information received through the movement determination unit **307** and the actual viewpoint position and orientation information about the user received from the actual viewpoint acquisition unit **309**. In this case, the virtual viewpoint generation unit **310** calculates the movement amount of the virtual viewpoint of the user based on the operation information received from the movement determination unit **307**. Further, the virtual viewpoint generation unit **310** adds the calculated virtual viewpoint movement amount of the user to the actual viewpoint position and orientation of the user received from the actual viewpoint acquisition unit **309**, thereby calculating the virtual viewpoint position and orientation in world coordinates. For example, if walk-through or the like is executed in the apparatus **301**, the virtual viewpoint generation unit **310** calculates the movement amount of the virtual viewpoint along with the walk-through or the like of the user, and adds the calculated movement amount to the actual viewpoint position and orientation, thereby calculating the virtual viewpoint position and orientation moved by the walk-through or the like. Various existing methods may be used to calculate the virtual viewpoint movement amount based on the operation information, and descriptions thereof are omitted.

(41) If the operation unit **305** and the actual viewpoint acquisition unit **309** are configured to acquire the position and orientation of body parts such as hands of the user, the position and orientation of the body parts are acquired and the viewpoint movement amount is also added to the position and orientation of the body parts. The position and orientation of the body parts such as hands of the user can be utilized for, for example, representation of the body parts in the avatar

whose display position is moved based on the viewpoint movement.

(42) In step **S440**, the transmission unit **315** transmits the actual viewpoint position and orientation and the virtual viewpoint position and orientation of the user to the apparatus **318**, and the reception unit **314** receives the actual viewpoint position and orientation and the virtual viewpoint position and orientation of the other user that are acquired and calculated in the same manner as described above in the apparatus **318** and are transmitted from the apparatus **318**. The actual viewpoint position and orientation and the virtual viewpoint position and orientation of the both users are thereby shared between the apparatus **301** and the apparatus **318**.

(43) If the apparatus **301** and the apparatus **318** have acquired the position and orientation of the body parts of the both users, the apparatus **301** and the apparatus **318** share the position and orientation information about the body parts of the both users by transmitting and receiving the position and orientation information.

(44) In step **S450**, the relative relationship calculation unit **313** and the correction unit **312** check if the movement flag in the operation information is ON. If the movement flag is ON (YES in step **S450**), the processing proceeds to step **S460**. In contrast, if the movement flag is not ON (NO in step **S450**), the virtual viewpoint position and orientation information about the user generated by the virtual viewpoint generation unit **310** is sent from the relative relationship calculation unit **313** to the combining unit **311** through the correction unit **312**. The processing of the information processing apparatus **319** then proceeds to step **S480**.

(45) In step **S460**, the relative relationship calculation unit **313** calculates a relative position and orientation between the shared actual viewpoint position and orientation of the other user and the actual viewpoint position and orientation of the user. In the present exemplary embodiment, the relative relationship calculation unit **313** calculates the actual viewpoint relative position and orientation based on the actual viewpoint position and orientation information about the user received through the virtual viewpoint generation unit **310** and the actual viewpoint position and orientation information about the other user received by the reception unit **314**.

(46) If the position and orientation of the body parts are also shared between the apparatus **301** and the apparatus **318**, the relative relationship calculation unit **313** also calculates the relative position and orientation between the body parts of the user and the body parts of the other user.

(47) In step **S470**, the correction unit **312** performs processing for correcting a mismatch between the virtual viewpoint of the other user and the virtual viewpoint of the user. In the present exemplary embodiment, the correction unit **312** performs correction processing to move the virtual viewpoint position and orientation of the user to match the virtual viewpoint position and orientation of the other user by walk-through in the apparatus **318**. In this case, the correction unit **312** performs correction processing to move the virtual viewpoint of the user to the coordinate system based on the virtual viewpoint of the other use. Specifically, the correction unit **312** performs correction processing to move the virtual viewpoint of the user to the origin of the coordinate system based on the virtual viewpoint of the other user. In other words, the correction unit **312** performs correction processing to move the virtual viewpoint of the user by adding the actual viewpoint relative position and orientation calculated by the relative relationship calculation unit **313** to the virtual viewpoint position and orientation of the other user. More specifically, the correction unit **312** adds the actual viewpoint relative position and orientation to the virtual viewpoint coordinate system with the origin corresponding to the virtual viewpoint position and orientation of the other user, thereby calculating the relative position and orientation corresponding to the virtual viewpoint of the user in the virtual viewpoint coordinate system of the other user. In this case, the virtual viewpoint position and orientation of the user are moved to the coordinate system with the origin corresponding to the virtual viewpoint of the other user in a state where the actual viewpoint relative position and orientation between the both users are maintained.

Accordingly, as a motion in the world coordinate system, the actual viewpoint position and the virtual viewpoint position of the other user overlap, and the virtual viewpoint position and

orientation based on the world coordinate system are moved by the movement amount.

(48) In step **S480**, the combining unit **311** generates a 3D image based on the virtual viewpoint position and orientation of the user, and combines the generated 3D image with the actual environment captured image by superimposing the 3D image on the actual environment image, thereby generating a mixed reality image as a combination of the virtual image and the actual environment image. For example, if the movement flag is ON, the mixed reality image corresponds to the image that can be seen after correction processing is performed to match the virtual viewpoint position and orientation of the other user by walk-through. For example, if the movement flag is OFF and walk-through or the like is executed in the apparatus **301**, the mixed reality image corresponds to the image that can be seen in the virtual viewpoint position and orientation of the user by walk-through or the like. In the present exemplary embodiment, the combining unit **311** also generates an avatar of the user based on the virtual viewpoint position and orientation, and combines the generated avatar with the actual environment image. Thus, the mixed reality image includes the avatar. An avatar of the other user may be generated using the shared information about the virtual viewpoint position and orientation of the other user. The avatar may represent, for example, only the head of the user, or may include the body parts, such as the torso and legs, located at positions estimated based on the head position.

(49) FIG. 5 illustrates an example of the mixed reality space to be observed in the virtual viewpoint position and orientation of the user when the other user executes walk-through to execute the information processing according to the present exemplary embodiment illustrated in the flowchart of FIG. 4. FIG. 5 corresponds to the example illustrated in FIG. 2. The user corresponds to the user **101** and the other user corresponds to the user **102**.

(50) As described above, the correction unit **312** performs the processing for correcting the virtual viewpoint position and orientation calculated by adding the actual viewpoint relative position and orientation between the user **101** and the user **102** (other user) to the virtual viewpoint position and orientation of the user **102**, to thereby obtain the corrected virtual viewpoint position and orientation of the user **101**.

(51) The virtual viewpoint position of the user **101** is thereby moved to the virtual viewpoint coordinate system of the user **102** in a state where the relative positional relationship between the actual viewpoint of the user **102** and the actual viewpoint of the user **101** is maintained. In the example illustrated in FIG. 2, the virtual viewpoint of the user **102** is moved to the position and orientation corresponding to the virtual viewpoint **104** by walk-through. As described above, the virtual viewpoint **104** of the user **102** is located at the intermediate position between the virtual model **105** and the virtual model **106** on the opposite side of the virtual model **105** and the virtual model **106**. Accordingly, the view from the virtual viewpoint of the user **101** after the correction processing as illustrated in FIG. 3 described above is updated with the view illustrated in FIG. 5. Specifically, the view of each object located in the mixed reality space is changed such that the virtual model **105** that is seen on the left side from the user **101** in FIG. 2 is seen on the back side and substantially in front of the user **101** in FIG. 5. The actual viewpoint position of the user **102** overlaps the position of the virtual viewpoint **104**, and the avatar **103** of the user **102** is seen at the entity position of the user **102**. The virtual model **106** that is seen on the right back side in FIG. 3 is seen on the left side in FIG. 5.

(52) In the case of providing the mixed reality to the user and a plurality of other users, the information processing apparatus **319** acquires the actual viewpoint position and orientation and the virtual viewpoint position and orientation of each of the other users from the information processing apparatus of each of the other users. The information processing apparatus **319** calculates the actual viewpoint relative position and orientation between the actual viewpoint position and orientation of each of the other users and the actual viewpoint position and orientation of the user, and corrects the virtual viewpoint position and orientation of the user based on, for example, the actual viewpoint relative position and orientation calculated for a selected desired

other user. In the case of selecting the desired other user, the desired other user may be selected by the user through the operation unit **305**, or may be automatically selected based on the positional relationship between the other user and the user in the actual space. For example, if the other user located near the user in the actual space executes walk-through, the mismatch in the relationship between the virtual viewpoint of the user and the virtual viewpoint of the other user by walk-through is likely to be conspicuous. Accordingly, it may be desirable to select, for example, the other user located within a predetermined distance from the user.

(53) While the flowchart of FIG. **4** illustrates an example where the virtual viewpoint position of the user is moved while the actual viewpoint relative position and orientation between the user and the other user are maintained, the virtual viewpoint position of the user may be moved to the same position and orientation as the virtual viewpoint position of the other user. In this case, the way how the user can see the virtual space after the movement is the same as the way how the other user can see the virtual space after the movement.

(54) The above-described exemplary embodiment illustrates an example where the virtual viewpoint position and orientation of the user is corrected to match the virtual viewpoint position and orientation of the other user based on the actual viewpoint relative position and orientation between the both users and the virtual viewpoint position and orientation of the other user. As a modified example, in the present exemplary embodiment, the virtual viewpoint position and orientation of the user may be arbitrarily moved by the user while the actual viewpoint relative position and orientation between the user and the other user are maintained in the mixed reality space. In this case, the view of each object located in the mixed reality space corresponds to the view from the virtual viewpoint position and orientation of the user. In this case, the position of the avatar displayed based on the virtual viewpoint position and orientation of the other user is updated along with the movement of the virtual viewpoint position and orientation of the user. In other words, when the virtual viewpoint position and orientation of the user are moved through the operation on the operation unit **305** such that the position of the avatar of the other user matches the entity position of the other user, the view seen by the user corresponds to the view illustrated in the example of FIG. **5**.

(55) As described above, the information processing apparatus **319** according to the present exemplary embodiment calculates the virtual viewpoint position of the user in the virtual viewpoint coordinate system with the origin corresponding to the virtual viewpoint of the other user during walk-through, based on the virtual viewpoint position and orientation information and the actual viewpoint relative position and orientation information obtained by walk-through. Specifically, in the information processing apparatus **319** according to the present exemplary embodiment, correction processing is performed to move the virtual viewpoint position and orientation of the user based on the position and orientation of the other user by walk-through and the actual viewpoint position and orientation of the user, thereby reducing a positional deviation between the virtual viewpoints of the both users. Consequently, during walk-through in the MR or AR executed by a plurality of users, the mismatch in the relationship between the virtual viewpoint and the actual viewpoint of each user during walk-through can be eliminated, so that the users are less likely to be confused in communication.

(56) In the first exemplary embodiment described above, after the movement flag is turned on upon detection of the operation input of the movement trigger and the virtual viewpoint position is corrected, the virtual viewpoint of each user is moved based on the operation information about each user in each apparatus until the next movement trigger is detected. A second exemplary embodiment illustrates an example where when the other user continuously moves the virtual viewpoint by walk-through, the virtual viewpoint position and orientation of the user are continuously corrected along with the movement of the virtual viewpoint position and orientation of the other user during the walk-through.

(57) FIG. **6** is a block diagram illustrating a configuration example of a system including an

information processing apparatus **319** according to the second exemplary embodiment. In the configuration example illustrated in FIG. **6**, the functional units similar to the functional units illustrated in FIG. **1** are denoted by the same reference numerals, and descriptions thereof are omitted. Configurations and processing that are different from those of the first exemplary embodiment will be described below.

(58) The information processing apparatus **319** of the apparatus **301** illustrated in FIG. **6** includes a continuation determination unit **601**. The continuation determination unit **601** includes not only the function similar to that of the movement determination unit **307** described above, but also a function for determining the presence or absence of predetermined movement continuation information indicating an input of a virtual viewpoint movement continuation operation in the operation information received from the operation acquisition unit **306**. In other words, the continuation determination unit **601** functions as a continuation detection unit to detect the predetermined movement continuation information. Upon detecting the predetermined movement continuation information, the continuation determination unit **601** outputs, for example, the operation information in which a continuation flag is ON. If the continuation flag is ON, the relative relationship calculation unit **313** and the correction unit **312** continuously perform the above-described processing.

(59) FIG. **7** is a flowchart illustrating a processing flow of information processing performed in the information processing apparatus **319** of the apparatus **301** according to the second exemplary embodiment. Similarly to the example illustrated in the flowchart of FIG. **4**, the processing in the flowchart of FIG. **7** is also performed in the information processing apparatus **319** of the apparatus **301** and the information processing apparatus of the apparatus **318**. In the flowchart of FIG. **7**, the same processing steps as those of FIG. **4** are denoted by the same step numbers as those in FIG. **4**, and descriptions thereof are omitted.

(60) In the flowchart of FIG. **7**, the processing of the information processing apparatus **319** proceeds to step **S710** after the processing of step **S410**. In step **S710**, the continuation determination unit **601** performs processing for detecting movement continuation information indicating that the movement to the virtual viewpoint of other user is continued based on the operation information received from the operation acquisition unit **306**. In step **S720**, the continuation determination unit **601** determines whether to continue the movement of the virtual viewpoint based on the detection result on the operation information. If the continuation determination unit **601** determines that the movement of the virtual viewpoint is continued (YES in step **S720**), the processing proceeds to step **S730**. In contrast, if the continuation determination unit **601** determines that the movement of the virtual viewpoint is not continued (NO in step **S720**), the processing proceeds to step **S420**. The processing of step **S420** and subsequent steps is similar to that in the flowchart of FIG. **4**, and thus descriptions thereof are omitted. However, after the processing of step **S440**, the processing proceeds to step **S460** in the second exemplary embodiment.

(61) In step **S730**, the relative relationship calculation unit **313** continuously calculates the viewpoint movement amount of the virtual viewpoint of the other user based on the virtual viewpoint position and orientation information about the other user received by the reception unit **314**, and sends the calculated viewpoint movement amount to the correction unit **312**.

(62) In step **S740**, the correction unit **312** continuously adds the viewpoint movement amount of the virtual viewpoint of the other user calculated in step **S730** to the virtual viewpoint position and orientation of the user in world coordinates, thereby calculating the virtual viewpoint position and orientation of the user. Accordingly, the virtual viewpoint position of the user can be moved by following the movement of the virtual viewpoint of the other user by walk-through. After the processing of step **S740**, the processing proceeds to step **S480**. In step **S480**, the mixed reality space image is generated in the same manner as described above.

(63) In the first and second exemplary embodiments described above, the movement trigger to be

input through the operation unit **305** is used to determine whether to move the virtual viewpoint of the user to match the virtual viewpoint of the other user. In contrast, a third exemplary embodiment illustrates an example where the viewpoint position and orientation of the user and the viewpoint position and orientation of the other user are used to determine whether to move the virtual viewpoint of the user to match the virtual viewpoint of the other user.

(64) An information processing apparatus **319** according to the third exemplary embodiment calculates the magnitude of a relative position and orientation (referred to as a virtual viewpoint relative position and orientation) between the virtual viewpoint of the user and the virtual viewpoint of the other user. The information processing apparatus **319** according to the third exemplary embodiment acquires, as the movement trigger to move the virtual viewpoint of the user, information indicating that the magnitude of the virtual viewpoint relative position and orientation is greater than a predetermined threshold for relative position and orientation. Specifically, when the magnitude of the virtual viewpoint relative position and orientation between the user and the other user is greater than the predetermined threshold for relative position and orientation, the information processing apparatus **319** according to the third exemplary embodiment executes processing for correcting a mismatch between the virtual viewpoint of the other user by walk-through and the virtual viewpoint of the user. The present exemplary embodiment assumes a case where the virtual viewpoint of the other user has moved beyond an expected movement range by walk-through as a comparison condition in which the magnitude of the virtual viewpoint relative position and orientation is greater than the threshold for relative position and orientation. Specifically, the present exemplary embodiment assumes a case where when the virtual viewpoint of the other user has moved beyond the expected movement range by walk-through, the user automatically follows the movement of the virtual viewpoint of the other user.

(65) The information processing apparatus **319** according to the third exemplary embodiment calculates a distance between the virtual viewpoint of the user and the virtual viewpoint of the other user as information indicating the magnitude of the relative position and orientation between the user and the other user, and uses a predetermined distance threshold as the threshold for relative position and orientation. When the information processing apparatus **319** acquires, as the movement trigger, information indicating that the calculated distance is greater than the predetermined distance threshold, similarly to the above-described exemplary embodiments, the processing of calculating the actual viewpoint relative position and orientation between the user and the other user is performed and the processing of correcting the virtual viewpoint position and orientation of the user is further performed.

(66) FIG. **8** is a block diagram illustrating a configuration example of a system including the information processing apparatus **319** according to the third exemplary embodiment. In the configuration example illustrated in FIG. **8**, the functional units similar to the functional units illustrated in FIG. **1** are denoted by the same reference numerals, and descriptions thereof are omitted. Only the configuration and processing that are different from those of the first exemplary embodiment will be described below.

(67) The information processing apparatus **319** of the apparatus **301** illustrated in FIG. **8** does not include the movement determination unit **307** and the continuation determination unit **601** described above. The operation information acquired by the operation acquisition unit **306** is sent to the virtual viewpoint generation unit **310**. In the same manner as described above, the virtual viewpoint generation unit **310** generates the virtual viewpoint position and orientation of the other user, and sends the virtual viewpoint position and orientation information about the user and the actual viewpoint position and orientation information received from the actual viewpoint acquisition unit **309** to the transmission unit **315**. In the third exemplary embodiment, the virtual viewpoint position and orientation information and the actual viewpoint position and orientation information about the user output from the virtual viewpoint generation unit **310** are also sent to a relative relationship comparison unit **801**. The relative relationship comparison unit **801** also

receives the virtual viewpoint position and orientation information and the actual viewpoint position and orientation information about the other user that are transmitted from the transmission unit **316** of the apparatus **318** and are received by the reception unit **314**.

(68) The relative relationship comparison unit **801** includes not only the function of the relative relationship calculation unit **313** described above, but also a function for comparing a threshold with the magnitude of the relative position and orientation between the virtual viewpoint of the user and the virtual viewpoint of the other user. The relative relationship comparison unit **801** calculates the distance between the virtual viewpoints as the magnitude of the relative position and orientation between the virtual viewpoints of the both users, based on the virtual viewpoint position and orientation information about the user and the virtual viewpoint position and orientation information about the other user received by the reception unit **314**. If the relative relationship comparison unit **801** determines that the calculated distance between the virtual viewpoints is greater than the distance threshold, the relative relationship comparison unit **801** acquires this determination result as the movement trigger. Upon detecting the movement trigger, the relative relationship comparison unit **801** thereafter performs actual viewpoint relative position and orientation calculation processing, i.e., processing for calculating the relative position and orientation between the actual viewpoint position and orientation of the user and the actual viewpoint position and orientation of the other user, similarly to the relative relationship calculation unit **313** according to the exemplary embodiments described above. If the calculated distance between the virtual viewpoints is smaller than or equal to the distance threshold, the relative relationship comparison unit **801** sends the virtual viewpoint position and orientation information about the user generated by the virtual viewpoint generation unit **310**, to the combining unit **311**.

(69) Upon receiving the actual viewpoint relative position and orientation information from the relative relationship comparison unit **801**, the correction unit **312** according to the third exemplary embodiment executes processing for correcting a mismatch between the virtual viewpoint of the other user by walk-through and the virtual viewpoint of the user, in the same manner as described above. Specifically, the correction unit **312** adds the actual viewpoint relative position and orientation between the user and the other user to the virtual viewpoint position and orientation of the other user, thereby correcting the virtual viewpoint position and orientation of the user to match the virtual viewpoint of the other user by walk-through or the like.

(70) The combining unit **311** according to the third exemplary embodiment generates a 3D image based on the received virtual viewpoint position and orientation, and combines the generated 3D image with the actual environment captured image acquired by the image acquisition unit **308** by superimposing the 3D image on the actual environment image, thereby generating a mixed reality image as a combination of the virtual image and the actual environment image. In the third exemplary embodiment, after the correction unit **312** executes the correction processing, the mixed reality image generated by the combining unit **311** corresponds to the image that can be seen in the virtual viewpoint position and orientation of the other user by walk-through. In contrast, if the relative relationship comparison unit **801** determines that the distance is smaller than or equal to the distance threshold, the mixed reality image generated by the combining unit **311** corresponds to the image that can be seen in the virtual viewpoint position and orientation along with the walk-through or the like of the user.

(71) FIG. **9** is a flowchart illustrating a processing flow of information processing performed in the information processing apparatus **319** of the apparatus **301** according to the third exemplary embodiment. Similarly to the example illustrated in the flowchart of FIG. **4**, the processing in the flowchart of FIG. **9** is also performed in the information processing apparatus **319** of the apparatus **301** and the information processing apparatus of the apparatus **318**. In the flowchart of FIG. **9**, the same processing steps as those of FIG. **4** are denoted by the same step numbers as those in FIG. **4**, and descriptions thereof are omitted.

(72) In the flowchart of FIG. **9**, after the processing of step **S410**, the processing proceeds to step

S430, and after the processing of step **S440**, the processing proceeds to step **S910**. In step **S910**, the relative relationship comparison unit **801** calculates the relative position and orientation between the virtual viewpoint of the other user and the virtual viewpoint of the user.

(73) In step **S920**, the relative relationship comparison unit **801** determines whether the magnitude of the virtual viewpoint relative position and orientation, which is the relative position and orientation between the virtual viewpoint of the user and the virtual viewpoint of the other user, is greater than the predetermined threshold for relative position and orientation that is set in advance. In the present exemplary embodiment, the relative relationship comparison unit **801** calculates the distance between the virtual viewpoint of the user and the virtual viewpoint of the other user, and determines whether the calculated distance is greater than the predetermined distance threshold. If it is determined that the magnitude of the virtual viewpoint relative position and orientation between the user and the other user is greater than the threshold for relative position and orientation (YES in step **S920**), the processing of the information processing apparatus **319** proceeds to step **S470**. If the magnitude of the virtual viewpoint relative position and orientation between the user and the other user is not greater than the threshold for relative position and orientation (NO in step **S920**), the processing proceeds to step **S480**. The processing of steps **S470** and **S480** is similar to that described above, and thus descriptions thereof are omitted.

(74) In the determination processing of step **S920**, it is determined whether the magnitude of the virtual viewpoint relative position and orientation is greater than the threshold for relative position and orientation. However, the determination processing is not limited to this example. For example, as to the determination processing of step **S920**, it can be determined whether the magnitude of the virtual viewpoint relative position and orientation is smaller than or equal to the threshold for relative position and orientation, or whether the distance between the virtual viewpoint of the user and the virtual viewpoint of the other user is smaller than or equal to the predetermined distance threshold. In this case, when the distance between the virtual viewpoint of the other user that is moved by walk-through and the virtual viewpoint of the user is smaller than or equal to the distance threshold, or when the virtual point of the other user that is moved by walk-through and the virtual viewpoint of the user are located at a distance smaller than or equal to the distance threshold, the viewpoint of the user can be automatically moved to match the virtual viewpoint of the other user.

(75) While the third exemplary embodiment illustrates an example where the presence or absence of the movement trigger is determined based on the distance between the virtual viewpoint of the user and the virtual viewpoint of the other user, the presence or absence of the movement trigger may be determined based on a distance between the actual viewpoint of the user and the actual viewpoint of the other user. In this case, the relative relationship comparison unit **801** calculates the distance between the actual viewpoint of the user and the actual viewpoint of the other user based on the actual viewpoint position and orientation information about the user and the actual viewpoint position and orientation information about the other user. The relative relationship comparison unit **801** then acquires, as the movement trigger, information indicating that it is determined that the distance is greater than the distance threshold. Upon detecting the movement trigger, the relative relationship comparison unit **801** performs processing for calculating the actual viewpoint relative position and orientation between the user and the other user.

(76) According to the third exemplary embodiment, a positional deviation between the virtual viewpoint of the user and the virtual viewpoint of the other user can be automatically reduced without using the trigger operation input used in the first or second exemplary embodiment. Consequently, also in the third exemplary embodiment, during walk-through in the MR or AR executed by a plurality of users, the mismatch in the relationship between the virtual viewpoint and the actual viewpoint of each user during walk-through can be eliminated, so that the users are less likely to be confused in communication.

(77) The information processing apparatus **319** according to the third exemplary embodiment may

include the movement determination unit **307** described in the first exemplary embodiment and the continuation determination unit **601** described in the second exemplary embodiment.

(78) In addition, the information processing apparatus **319** is prepared for each user in the first to third exemplary embodiments described above. Alternatively, a single information processing apparatus **319** can be used to perform the above-described information processing by communicating with the operation unit **305** and the terminal **302** of each user.

(79) FIG. **10** illustrates an example of a hardware configuration in which the information processing apparatuses **319** according to each exemplary embodiment described above can be implemented.

(80) As illustrated in FIG. **10**, a central processing unit (CPU) **1020** comprehensively controls devices connected via a bus **1010**. The CPU **1020** reads out and executes processing steps and programs stored in a read-only memory (ROM) **1030**. An operating system (OS), information processing programs according to the exemplary embodiments, device drivers, and the like are stored in the ROM **1030**, and are temporarily stored in a random access memory (RAM) **1040** and are executed, as needed, by the CPU **1020**. An input interface (I/F) **1050** is used for the information processing apparatus **319** according to the exemplary embodiments to input signals from external apparatuses (e.g., the image capturing unit **303** of the terminal **302**, the operation unit **305**, and the apparatus **318**) in any format that can be processed by the information processing apparatus **319** according to the exemplary embodiments. An output OF **1060** is used for external apparatuses (e.g., the display unit **304** of the terminal **302**, and the apparatus **318**) to output signals in any format that can be processed by the external apparatuses. These functional units are implemented by the CPU **1020** loading programs stored in the ROM **1030** into the RAM **1040** and executing the programs in accordance with the flowcharts according to the exemplary embodiments described above. For example, if hardware is configured as an alternative for software processing using the CPU **1020**, an arithmetic unit or a circuit corresponding to the processing of each functional unit described herein may be configured.

(81) Each of the functional units in the information processing apparatus **319** according to the exemplary embodiments described above may be implemented by hardware, or some of the functional units may be executed by software. If the functional units are executed by software, the software is executed using, for example, the CPU **1020**, the ROM **1030**, and the RAM **1040** illustrated in FIG. **10**.

Other Embodiments

(82) Embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(83) While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the present disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(84) This application claims the benefit of Japanese Patent Application No. 2022-015693, filed Feb. 3, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. An information processing apparatus comprising: a memory; and at least one processor in communication with the at least one processor, the memory storing instructions that, when executed by the at least one processor, cause the at least one processor to: acquire an actual viewpoint position and orientation of a first user and a virtual viewpoint position and orientation of the first user; acquire an actual viewpoint position and orientation of each of a plurality of second users and a virtual viewpoint position and orientation of each of the plurality of second users; calculate, for each of the plurality of second users, a relative position and orientation between the actual viewpoint position and orientation of the first user and the actual viewpoint position and orientation of the second user; and correct the virtual viewpoint position and orientation of the first user based on the virtual viewpoint position and orientation of the second user selected from among the plurality of second users and a corresponding relative position and orientation between the actual viewpoint position and orientation of the first user and the selected second user.
2. The information processing apparatus according to claim 1, wherein the virtual viewpoint position and orientation of the first user is generated based on operation information about a viewpoint position input from the first user and the actual viewpoint position and orientation of the first user.
3. The information processing apparatus according to claim 1, wherein the memory storing instructions that, when executed by the at least one processor, cause the processor to transmit the actual viewpoint position and orientation of the first user and the virtual viewpoint position and orientation of the first user to an information processing apparatus of a second user included in the plurality of second users, wherein the memory storing instructions that, when executed by the at least one processor, cause the processor to receive the actual viewpoint position and orientation of the selected second user and the virtual viewpoint position and orientation of the selected second user transmitted from the information processing apparatus of the second user, and wherein the actual viewpoint position and orientation of the first user and the virtual viewpoint position and orientation of the first user, and the actual viewpoint position and orientation and the virtual viewpoint position and orientation of the selected second user are shared between an information processing apparatus of the first user and the information processing apparatus of the second user.
4. The information processing apparatus according to claim 1, wherein a virtual viewpoint of the first user is moved to a coordinate system representing the virtual viewpoint position and orientation of the selected second user.
5. The information processing apparatus according to claim 4, wherein the virtual viewpoint of the first user is moved to an origin of the coordinate system representing the virtual viewpoint position and orientation of the selected second user.
6. The information processing apparatus according to claim 5, wherein the virtual viewpoint of the first user is moved by adding the relative position and orientation to the virtual viewpoint position and orientation of the selected second user.
7. The information processing apparatus according to claim 6, wherein a relative position and orientation is calculated corresponding to the virtual viewpoint of the first user in a virtual viewpoint coordinate system of the second user by adding the relative position and orientation to a virtual viewpoint coordinate system with an origin corresponding to the virtual viewpoint position

and orientation of the selected second user.

8. The information processing apparatus according to claim 1, wherein upon detection of a predetermined trigger, the relative position and orientation is calculated, and the virtual viewpoint position and orientation of the first user is corrected.

9. The information processing apparatus according to claim 1, wherein upon detection of predetermined continuation information, the relative position and orientation is calculated, and the virtual viewpoint position and orientation of the first user is corrected.

10. The information processing apparatus according to claim 9, wherein a viewpoint movement amount of a virtual viewpoint of the second user is continuously based on the virtual viewpoint position and orientation of the second user, and wherein the virtual viewpoint position and orientation of the first user is corrected by adding the viewpoint movement amount that has been continuously calculated to the virtual viewpoint position and orientation of the first user.

11. The information processing apparatus according to claim 1, wherein upon detection of a predetermined trigger based on a relative position and orientation between the virtual viewpoint position and orientation of the first user and the virtual viewpoint position and orientation of the selected second user, the relative position and orientation between the actual viewpoint position and orientation of the first user and the actual viewpoint position and orientation of the selected second user is calculated.

12. The information processing apparatus according to claim 11, wherein the predetermined trigger is detected based on a comparison between a predetermined threshold and a magnitude of the relative position and orientation between the virtual viewpoint position and orientation of the first user and the virtual viewpoint position and orientation of the selected second user.

13. The information processing apparatus according to claim 12, wherein the predetermined trigger is detected, in a case where the magnitude of the relative position and orientation between the virtual viewpoint position and orientation of the first user and the virtual viewpoint position and orientation of the selected second user is greater than the predetermined threshold.

14. The information processing apparatus according to claim 12, wherein the predetermined trigger is detected, in a case where the magnitude of the relative position and orientation between the virtual viewpoint position and orientation of the first user and the virtual viewpoint position and orientation of the selected second user is smaller than or equal to the predetermined threshold.

15. The information processing apparatus according to claim 1, wherein upon detection of a predetermined trigger based on a comparison between a predetermined distance threshold and a distance between an actual viewpoint position of the first user and an actual viewpoint position of the second user, the relative position and orientation between the actual viewpoint position and orientation of the first user and the actual viewpoint position and orientation of the selected second user is calculated.

16. The information processing apparatus according to claim 15, wherein the predetermined trigger, in a case where the distance between the actual viewpoint position of the first user and the actual viewpoint position of the selected second user is greater than the predetermined distance threshold.

17. An information processing method comprising: acquiring, as a first acquisition, an actual viewpoint position and orientation of a first user and a virtual viewpoint position and orientation of the first user; acquiring, as a second acquisition, an actual viewpoint position and orientation of each of a plurality of second users and a virtual viewpoint position and orientation of each of the plurality of second users; calculating, for each of the plurality of second users, a relative position and orientation between the actual viewpoint position and orientation of the first user and the actual viewpoint position and orientation of the second user; and correcting the virtual viewpoint position and orientation of the first user based on the virtual viewpoint position and orientation of a second user selected from among the plurality of second users and a corresponding relative position and orientation between the actual viewpoint position and orientation of the first user and the actual viewpoint position and orientation of the selected second user.

18. A non-transitory computer-readable storage medium storing a program that causes a computer to execute an information processing method, the information processing method comprising: acquiring, as a first acquisition, an actual viewpoint position and orientation of a first user and a virtual viewpoint position and orientation of the first user; acquiring, as a second acquisition, an actual viewpoint position and orientation of each of a plurality of second users and a virtual viewpoint position and orientation of a each of the plurality of second users; calculating, for each of the plurality of second users, a relative position and orientation between the actual viewpoint position and orientation of the first user and the actual viewpoint position and orientation of the second user; and correcting the virtual viewpoint position and orientation of the first user based on the virtual viewpoint position and orientation of a second user selected from among the plurality of second users and a corresponding relative position and orientation between the actual viewpoint position and orientation of the first user and the selected second user.
